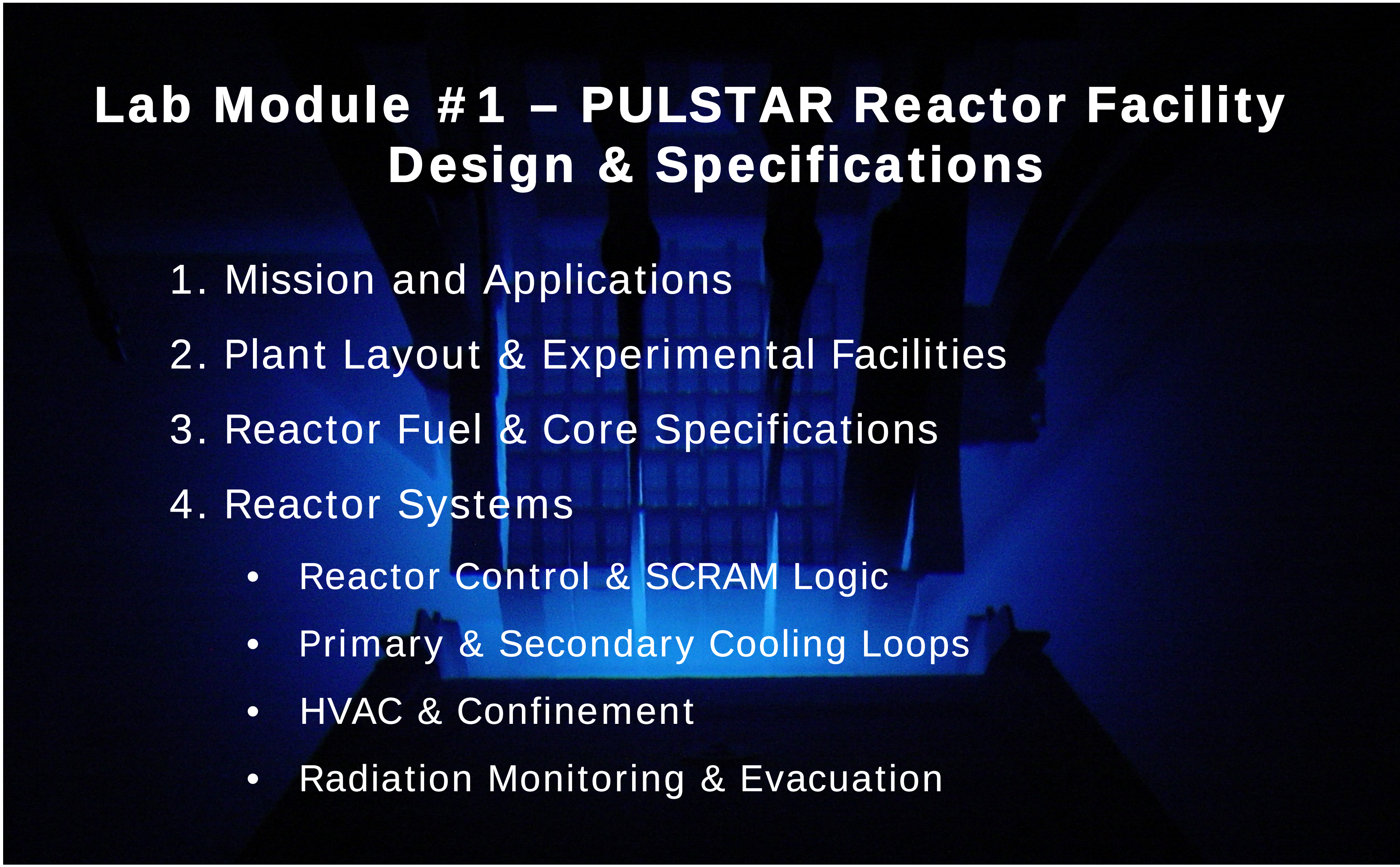
NE235 – Nuclear Reactor Operations Training



Lab Module #1 – PULSTAR Reactor Facility Design & Specifications

1. Mission and Applications
2. Plant Layout & Experimental Facilities
3. Reactor Fuel & Core Specifications
4. Reactor Systems
    - Reactor Control & SCRAM Logic
    - Primary & Secondary Cooling Loops
    - HVAC & Confinement
    - Radiation Monitoring & Evacuation

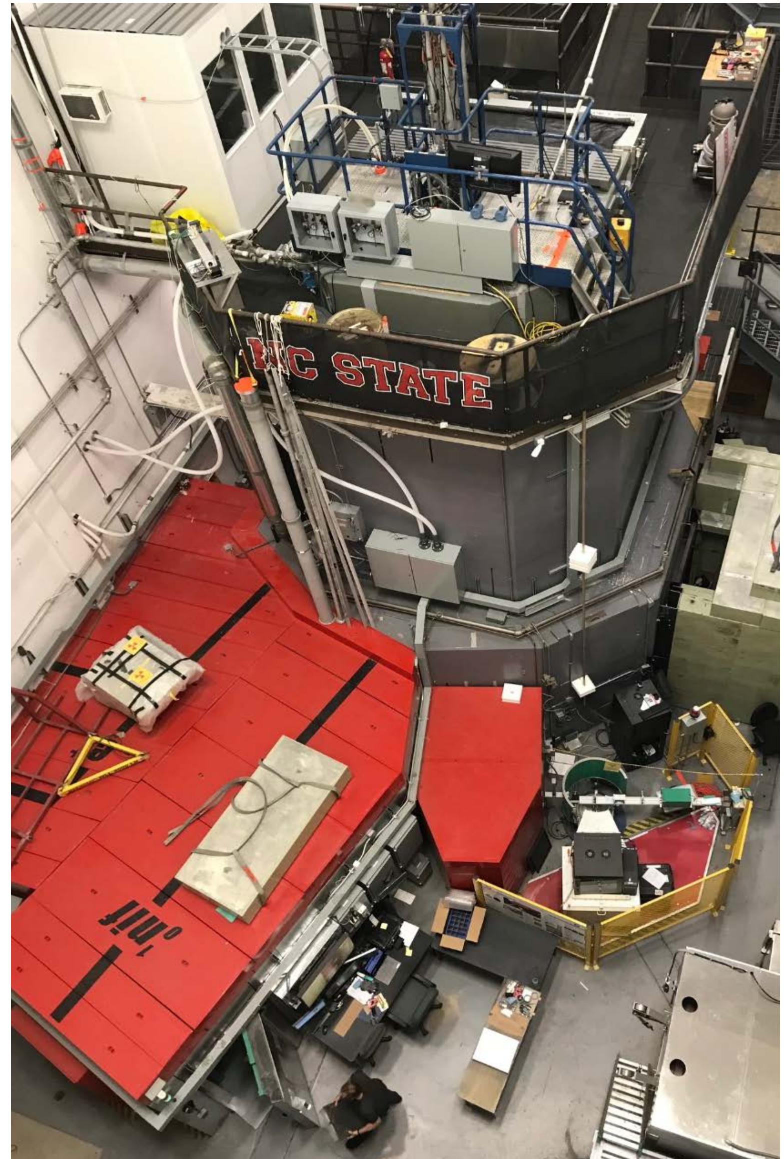
PULSTAR Reactor – Mission & Applications

Mission:

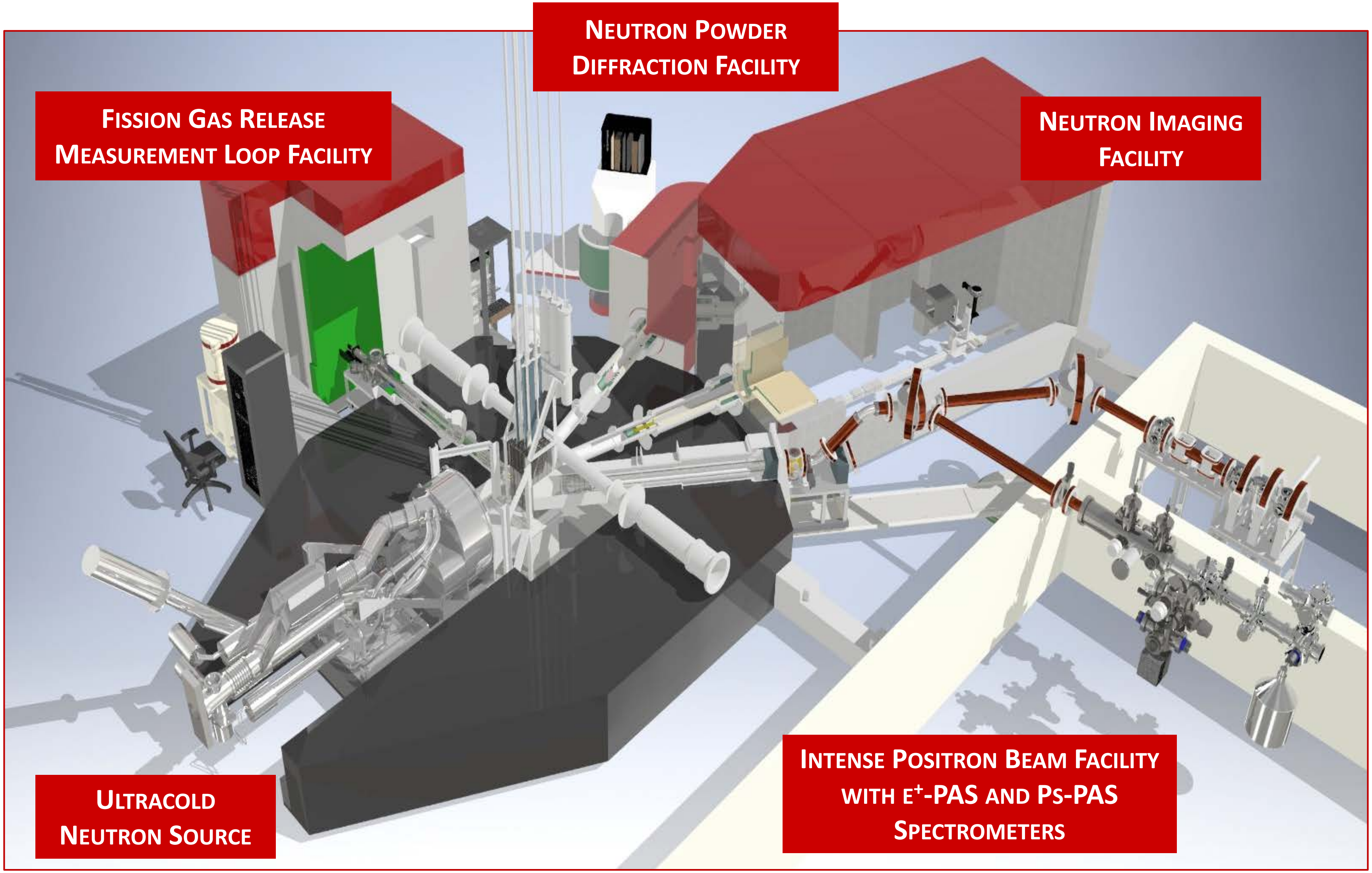
- Education, Research, Training, Service

Applications:

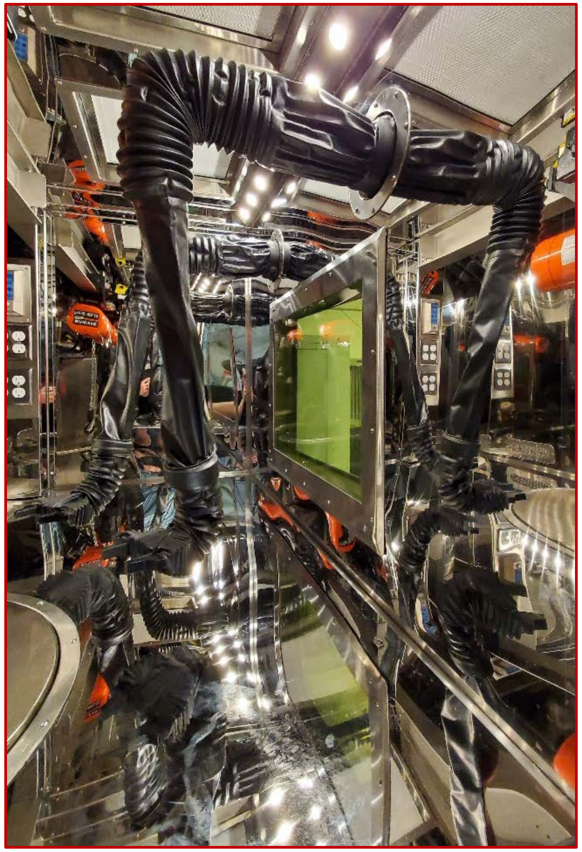
- Academic Laboratories
- Reactor Operator Training
- Distance Learning – Teleconferencing Labs
- Materials Research – E^+ Beam (IPB) with E+PALS/DBS spectrometers, Neutron Diffractometer (NDF), Fission Gas Release Measurement Loop (FGRML), Extreme Environment Irradiation Facility (EEIF)
- Fundamental Physics – UCN source
- Trace Element Analysis – environmental, forensic, industrial
- Non-Destructive Testing – neutron imaging (NIF)
- Detector Calibration – Navy, Power reactors
- Isotope Production – ^{166}Ho; ^{99}Mo



PULSTAR Reactor – Experimental Facilities



New Facilities & Upgrades

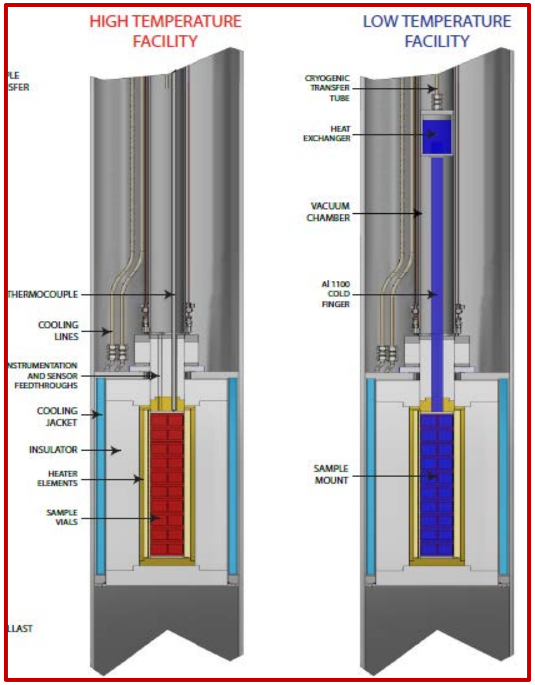


New Facilities:

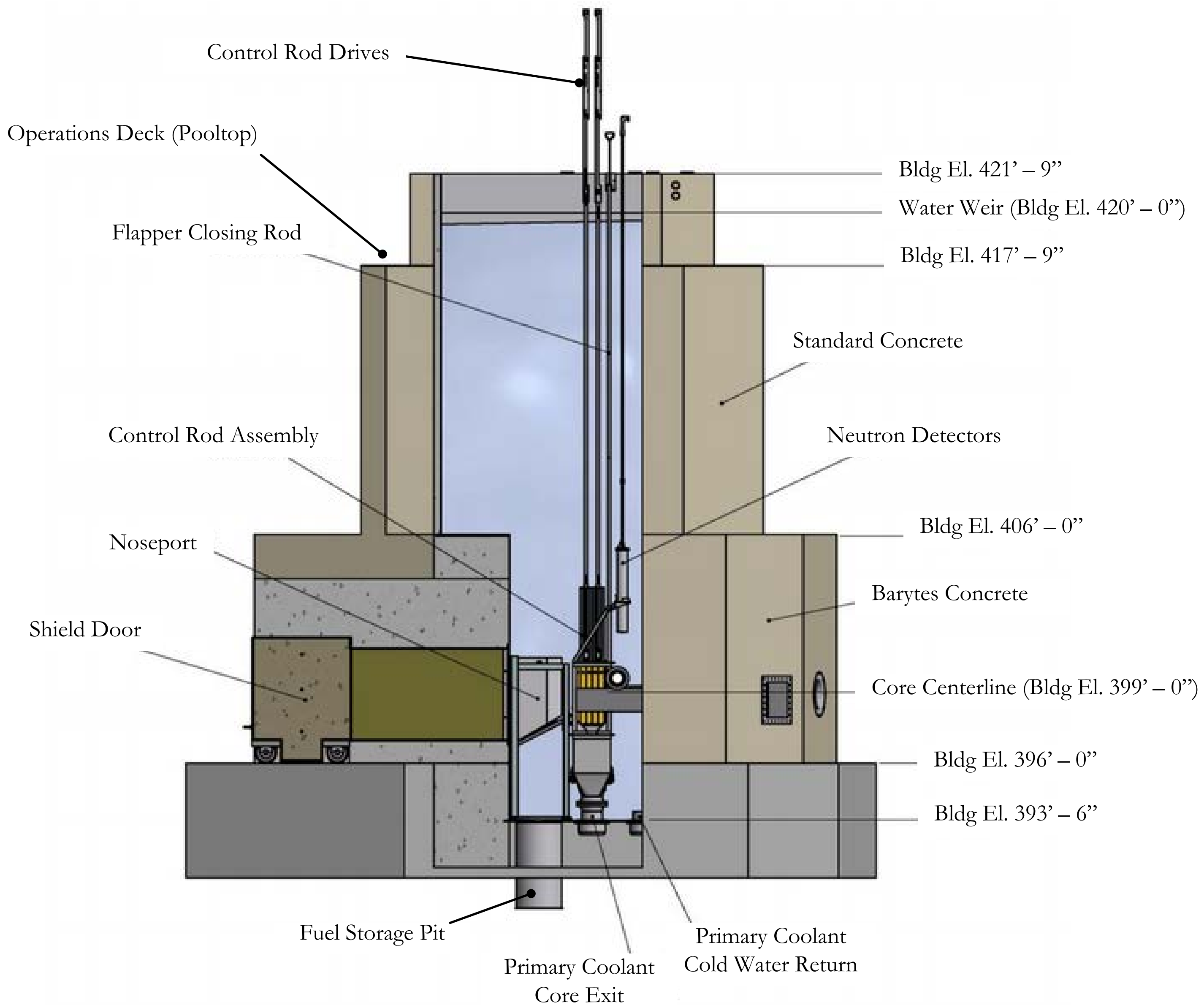
- **Hot Cell** – *isolated environment with 6" lead shielding and mating pool transfer cask*
- **Extreme Environment Irradiation Facility (EEIF):** *-200ºC to +800ºC for advanced reactor R&D (sensors, MSR materials).*

Upgrades:

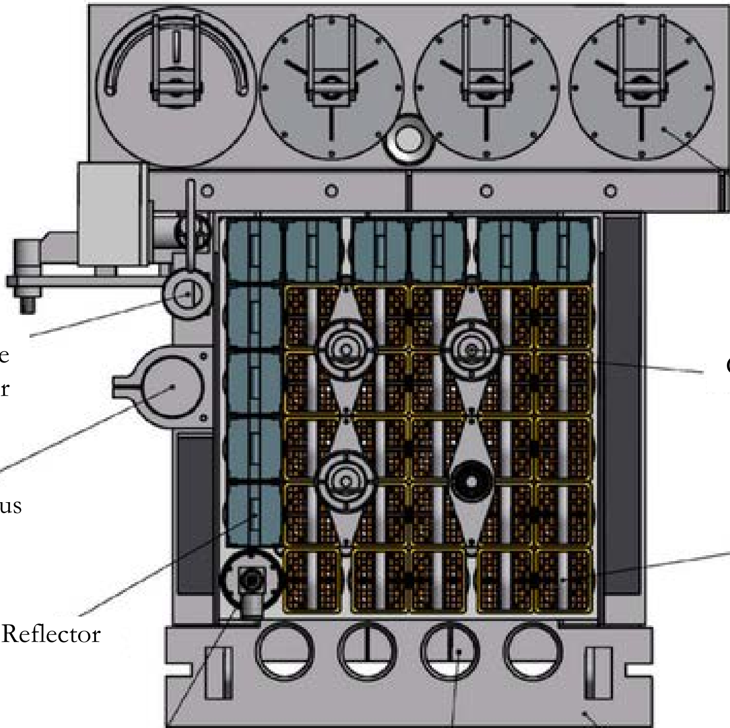
- **2 MW Power Upgrade:** *license approval pending with NRC*
- **E+ Beam PALS/DBS upgrades:** *Temperature stages; increased spatial and timing resolution; coincidence DBS.*



Reactor Plant – Elevation View



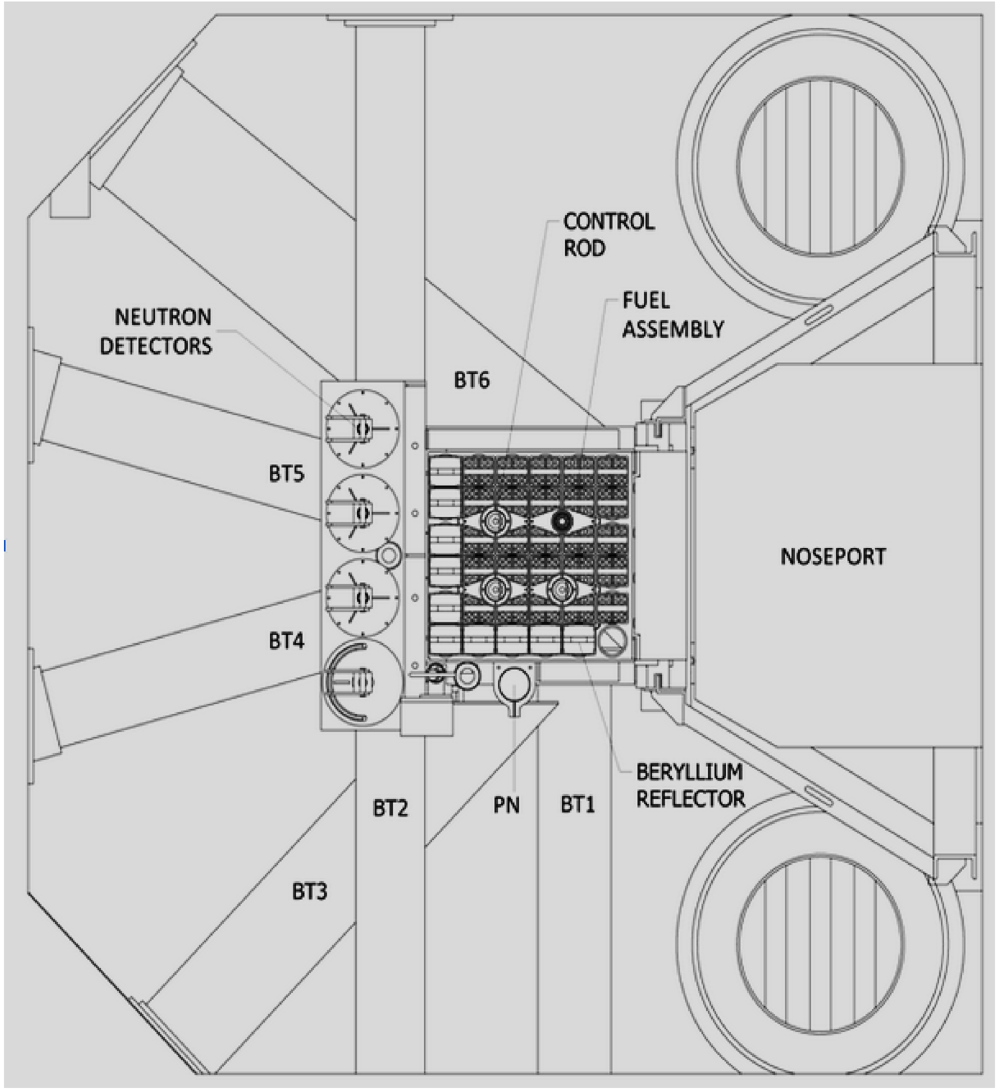
PULSTAR – Core and Gridplate Assembly



Core Structures:

- 36 Position Grid Plate
- 25 Fuel Assemblies
- 10 Reflector Elements
- 4 Control Rods
- 3 NI Power Monitors
- Cooling Plenum
- Flapper Valve Assembly
- Startup Source
- Fission Chamber

In-Pool Irradiation and Beam Port Facilities



Beam-Port Facilities:

- BP#1 – Fission Gas Measurement Loop Facility
- BP#4 - Neutron Diffractometer
- BP#5 - Neutron Imaging Facility
- BP#6 - Positron Intense Beam Fac.
- Noseport - UCN Source

In-Pool Irradiation Facilities:

- 4 Rotating Exposure Ports
- 2 Dry Tubes
- 1 PN Terminus (ex-core)
- 3.5" I.D. Standpipes

PULSTAR – Fuel Assembly

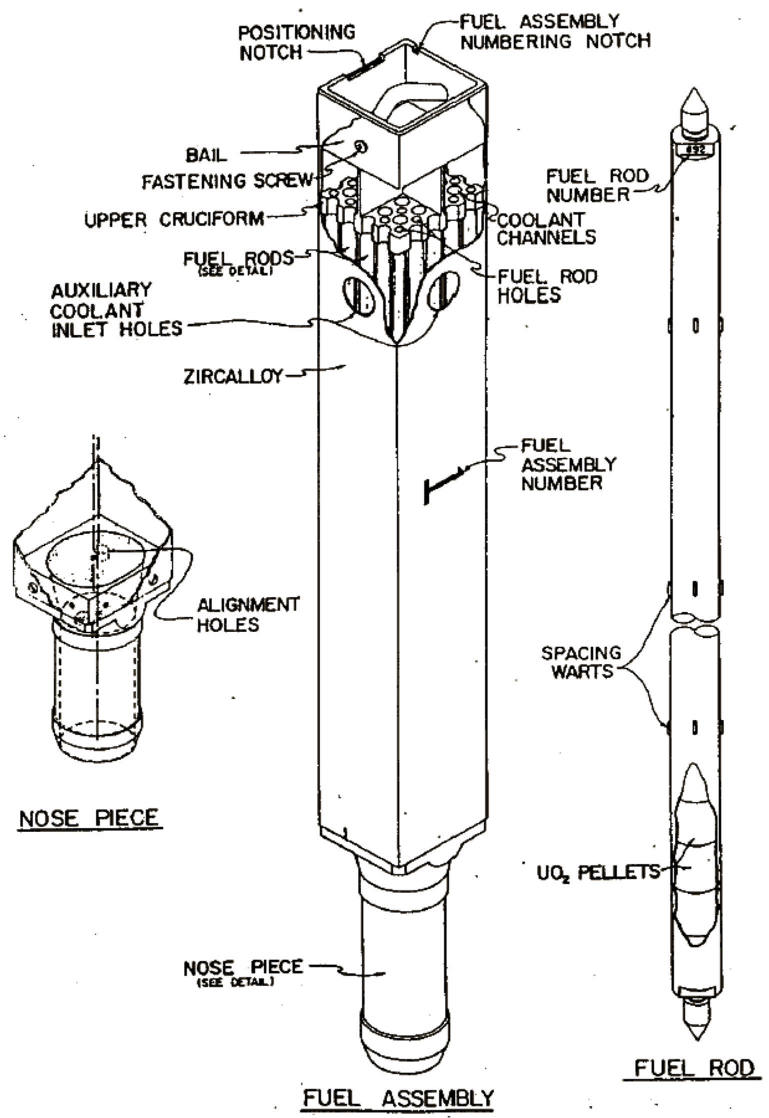


Figure 4.3 PULSTAR fuel

Fuel	
Material	UO_2
Form	Sintered Pellets
Enrichment (weight % U-235)	4%
Design Inventory Core (kg UO_2)	359
Density (gm/cm^3)	10.5 to 10.76
U-235 per Fuel Pin (gm)	20.02
Fuel Pin	
Pellet (diameter in (cm))	0.423 (1.074)(nominal)
Diametrical Gap (in (cm))	0.0085 (0.0216)(nominal)
Zircaloy-2 Clad Thickness (in (cm))	0.0205 (0.0521)(nominal)
Outside Diameter Pin (in (cm))	0.4725 (1.2002)(nominal)
Rectangular Spacing (center-to-center, in (cm))	0.606 x 0.524 (1.54 x 1.33)
Clearance (pin-to-pin, in (cm))	0.051 x 0.133 (0.130 x 0.338)
Clearance (pin-to-box, in (cm))	0.025 x 0.066 (0.064 x 0.168)
Height of Pellet Stack (in (cm))	24 (61)
Pins per Core	625
Height of Pellet (in (cm))	0.60 (1.52)
Fuel Box	
Material	Zr-2
Inside Dimensions (in (cm))	2.620 x 3.030 (6.655 x 7.700)
Wall Thickness (in (cm))	0.060 (0.152)
Clearance between Assemblies (in (cm))	0.040 (0.102)
Clearance between Control Rod Guide and Assemblies (in (cm))	0.060 (0.152)
Fuel Pins per Assembly	25
Weight (pounds (kg))	44 (20)

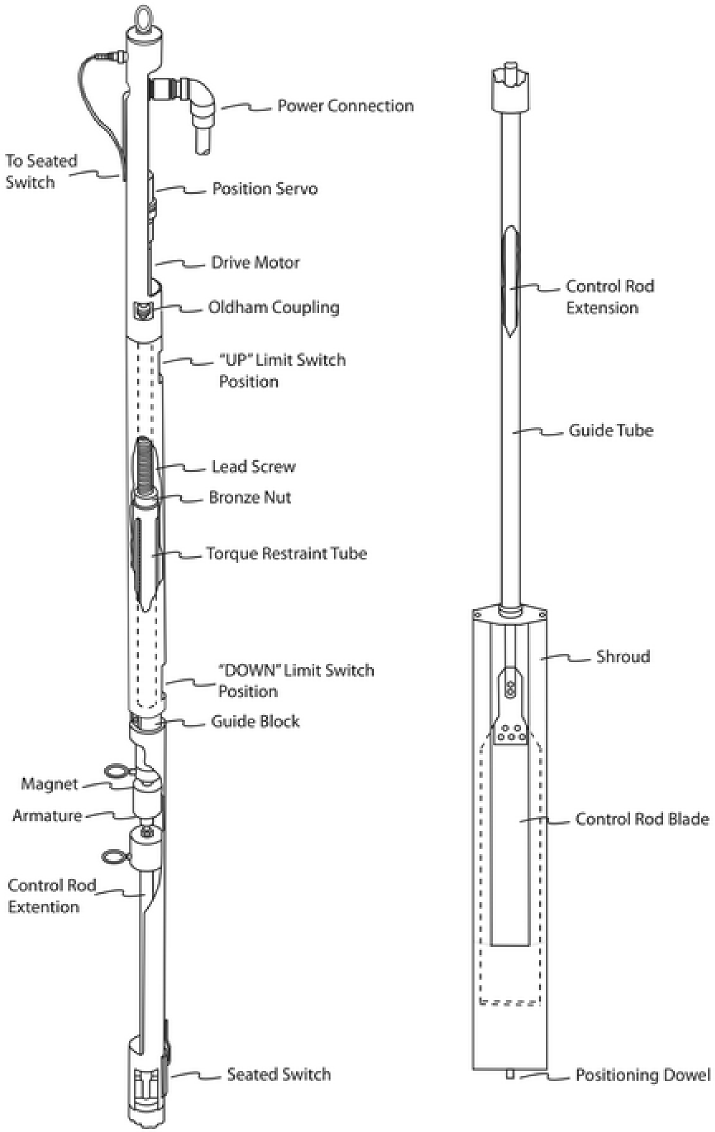
Reactor Core Specifications & Reactivity Coefficients

Core Dimensions	
Overall (in (cm))	15⅞ x 15 (40.3 x 38)
Height (in (cm))	24 (61)
Volume Fractions	
UO_2	0.3823
Gap (Helium)	0.0155
Cladding (includes warts)	0.0803
Lattice Water	0.3858
Assemblies (Zr-2)	0.0753
Water Between Assemblies	0.0255
Control Rod Guides (Al)	0.0128
Water Inside Guides (rods out)	0.0226
Total Volume Fractions	
UO_2	0.3823
Helium Gap	0.0155
H_2O	0.4339
Zr-2	0.1555
Aluminum	0.0128
Core Volume Ratios	
H_2/UO_2	1.135
H_2O/U	2.06

Minimum Critical Assemblies	20
Doppler Coefficient (pcm/°F)	-1.40
Void Coefficient (pcm/cm^3)	-1.03
Moderator Temperature Coefficient (pcm/°F)	-3.2
Beam Tube Worths (air to water-filled pcm)	
6 in (15.2 cm) diameter	193
8 in (20.3 cm) diameter	45
12 in x 12 in (30.5 cm x 30.5 cm)	20

Moderator, Reflector, and Coolant	
Material	Light Water
Nominal Inlet Temperature (°F (°C))	105 (40.6)
Nominal Outlet Temperature (°F (°C))	118.8 (48.2)
Primary Coolant Flow Rate (gpm (lps))	500 (31.5)
Secondary Coolant Flow Rate (gpm (lps))	700 (44.2)

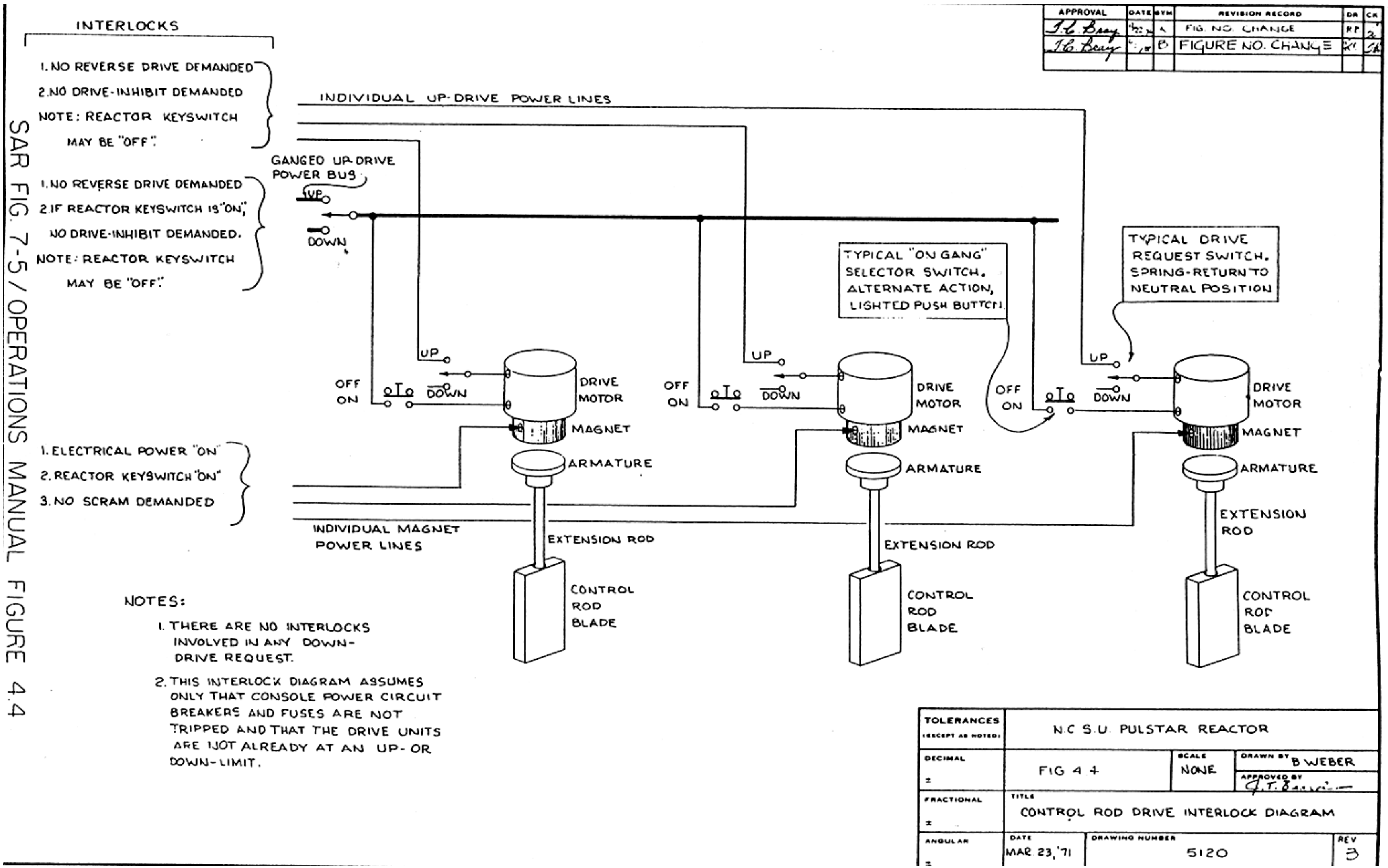
Control Rod Assembly



<u>Control Rods</u>

Absorber Material	Ag-In-Cd (80%-15%-5%)
Guide Material	Aluminum
Shape	Rectangular
Transverse Dimensions (in (cm))	
Guide	6.30 x 0.43 (16.0 x 1.09)
Absorber	4.85 x 0.18 (12.32 x 0.46)
Clearance Absorber to Guide (in (cm))	0.00625 (0.01588)
Clad Material	Sn/Ni
Number of Control Rods	3
Number of Non-Scrammable Rods	1

Control Rod – Electromagnet Assembly



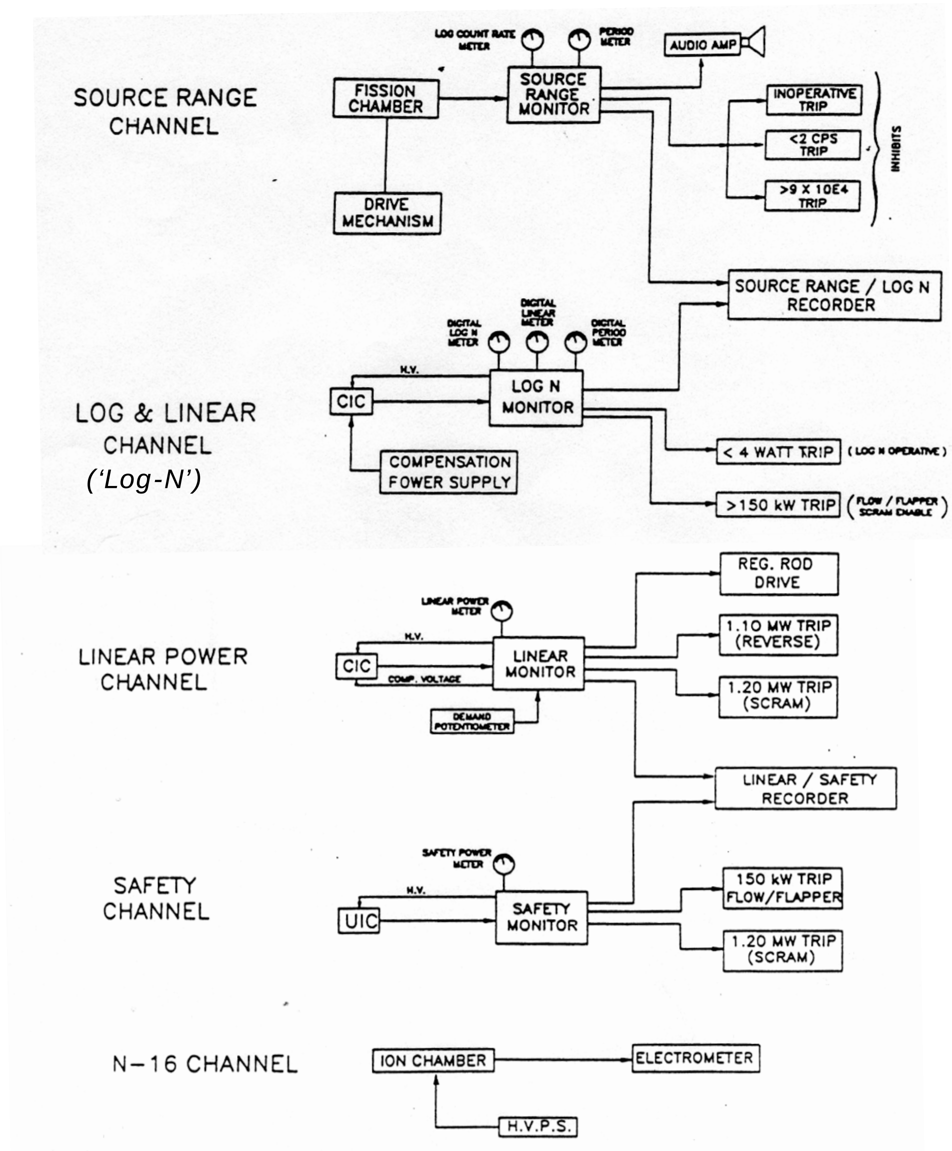
Reactor Instrumentation & SCRAM System

Seven Reactor SCRAM's:

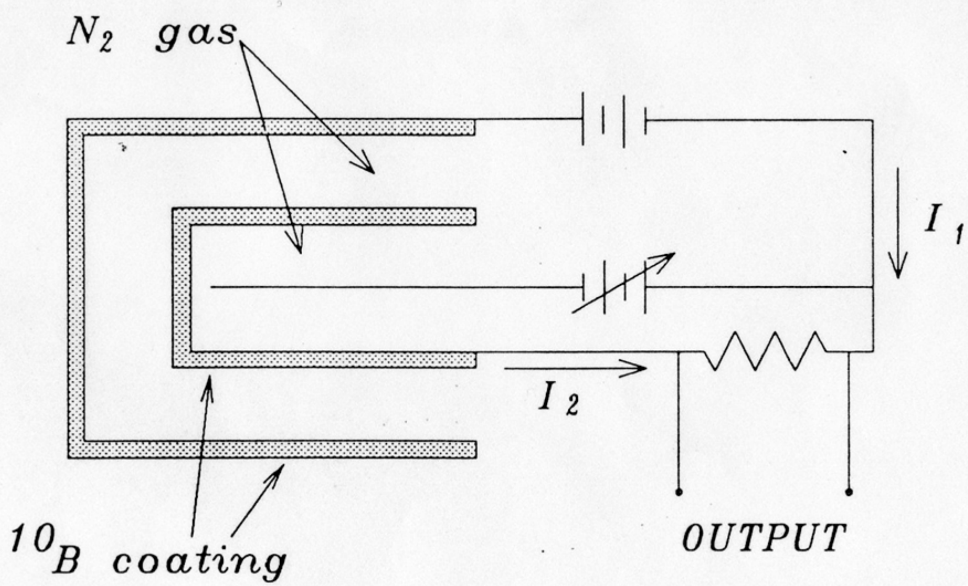
1. Linear Power Channel SCRAM at $\geq$ 120% Full Power (1MW)
2. Safety Power Channel SCRAM at $\geq$ 120% Full Power (1MW)
3. Primary Coolant Flow SCRAM at $\leq$ 475 GPM (*enables if Log-N or Safety Channel $\leq$ 150 kW*)
4. Flow Flapper Valve OPEN SCRAM (*enables if Log-N or Safety Channel $\leq$ 150 kW*)
5. Pool Level SCRAM at $\leq$ 36" below full
6. Pool Temperature SCRAM if T_2 (rtd) $\geq$ 116°F
7. Manual SCRAM – Depress Red Button on main console

Plus:

- Console Keyswitch
- SCRAM Logic Ground Fault



Nuclear Instrumentation – CIC Neutron Detector



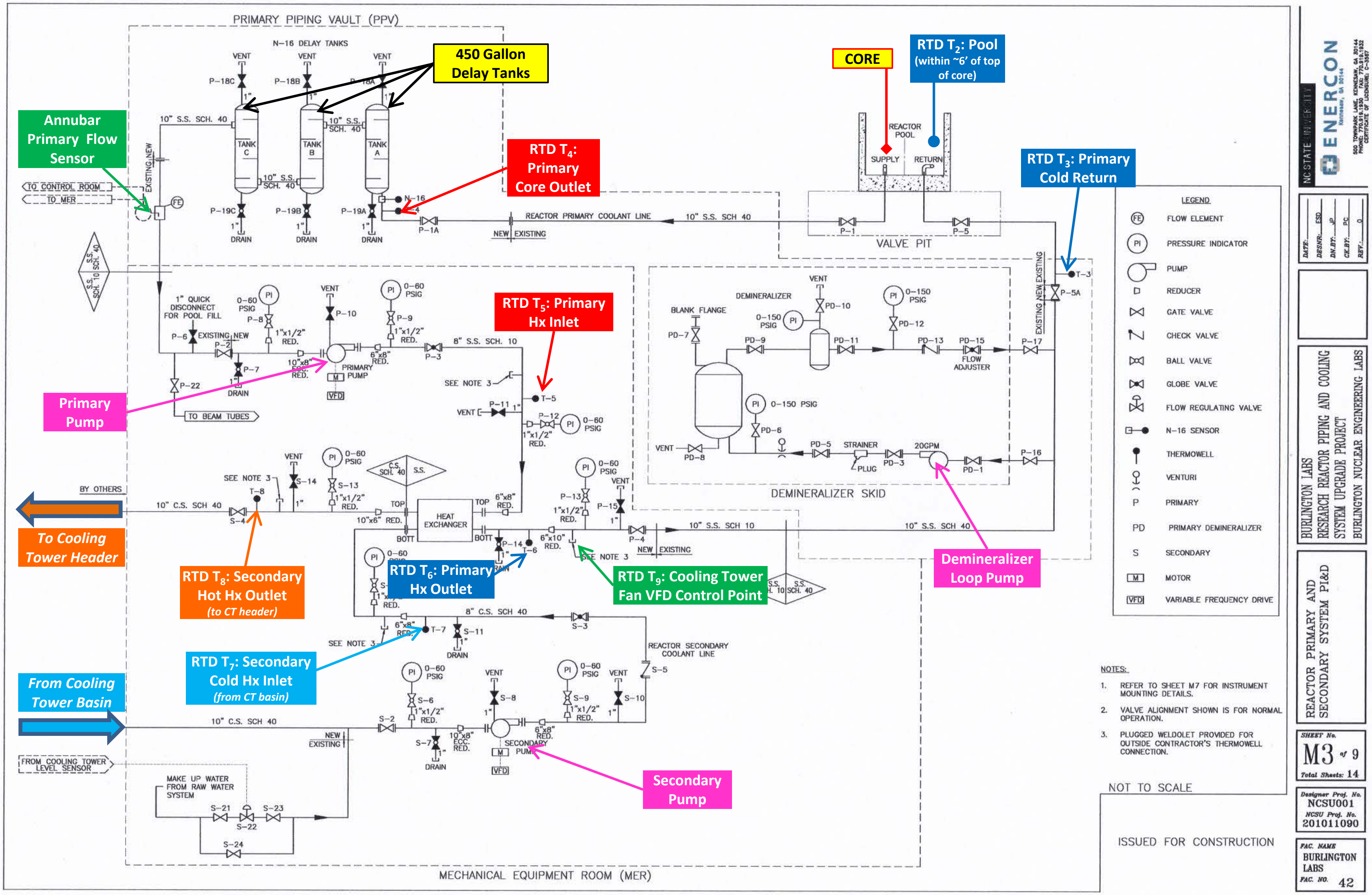
I_1 = Neutron and gamma current

I_2 = Gamma current

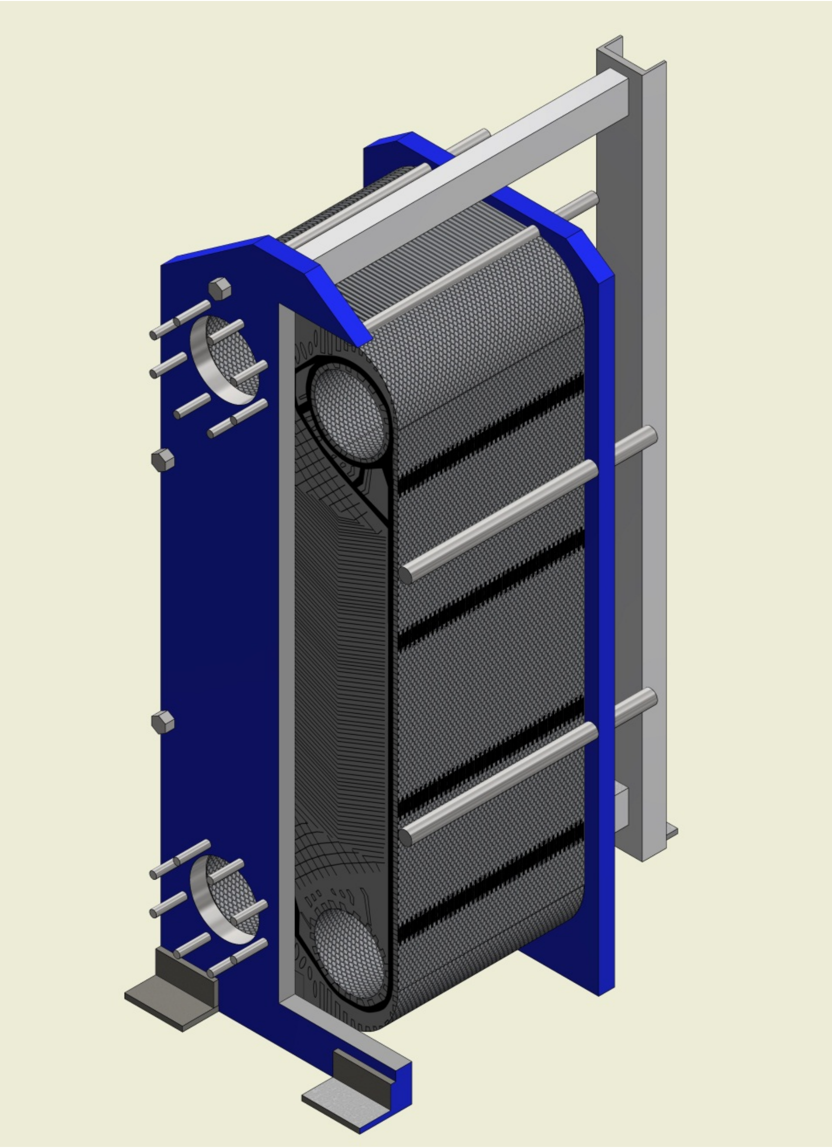
OUPUT CURRENT $= I_1 - I_2 =$ Neutron current

Figure 2.3 Compensated Ion Chamber

PULSTAR - Primary Cooling System Schematic



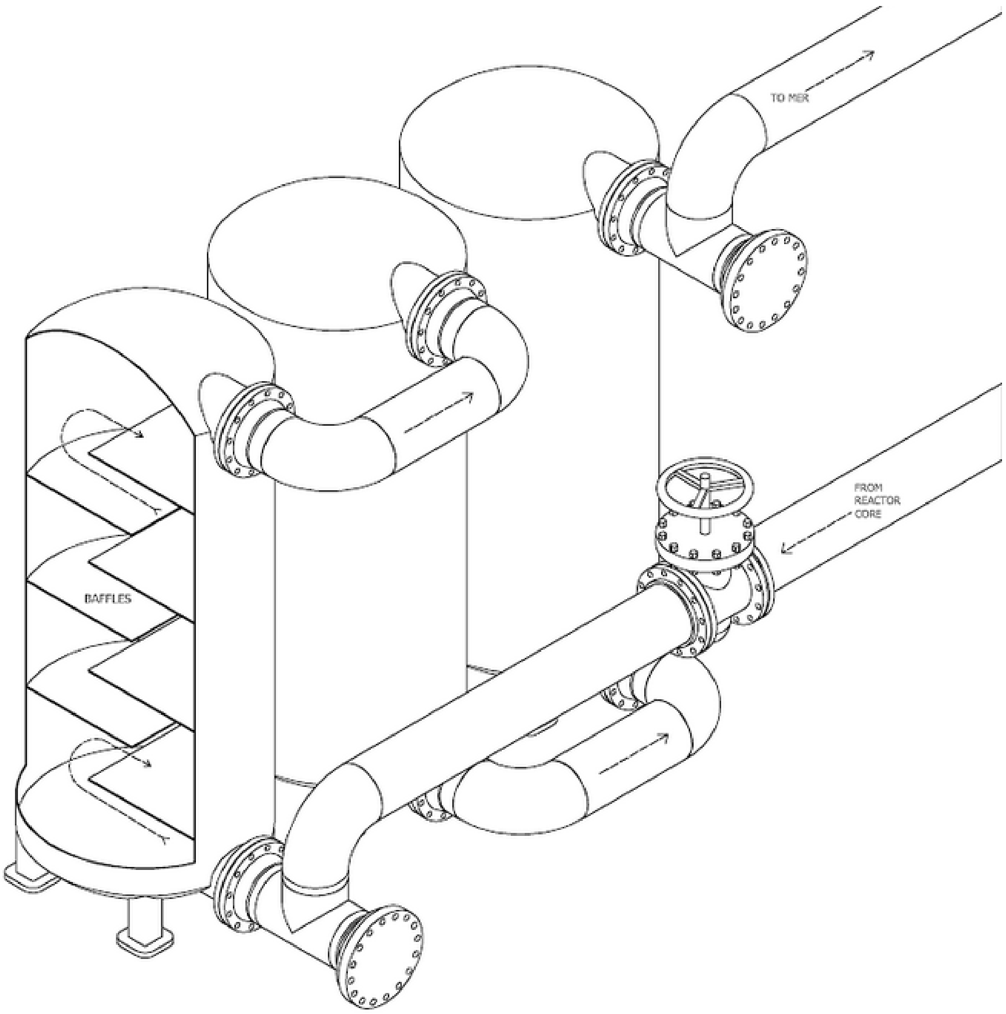
Alfa Laval Plate and Frame Heat Exchanger



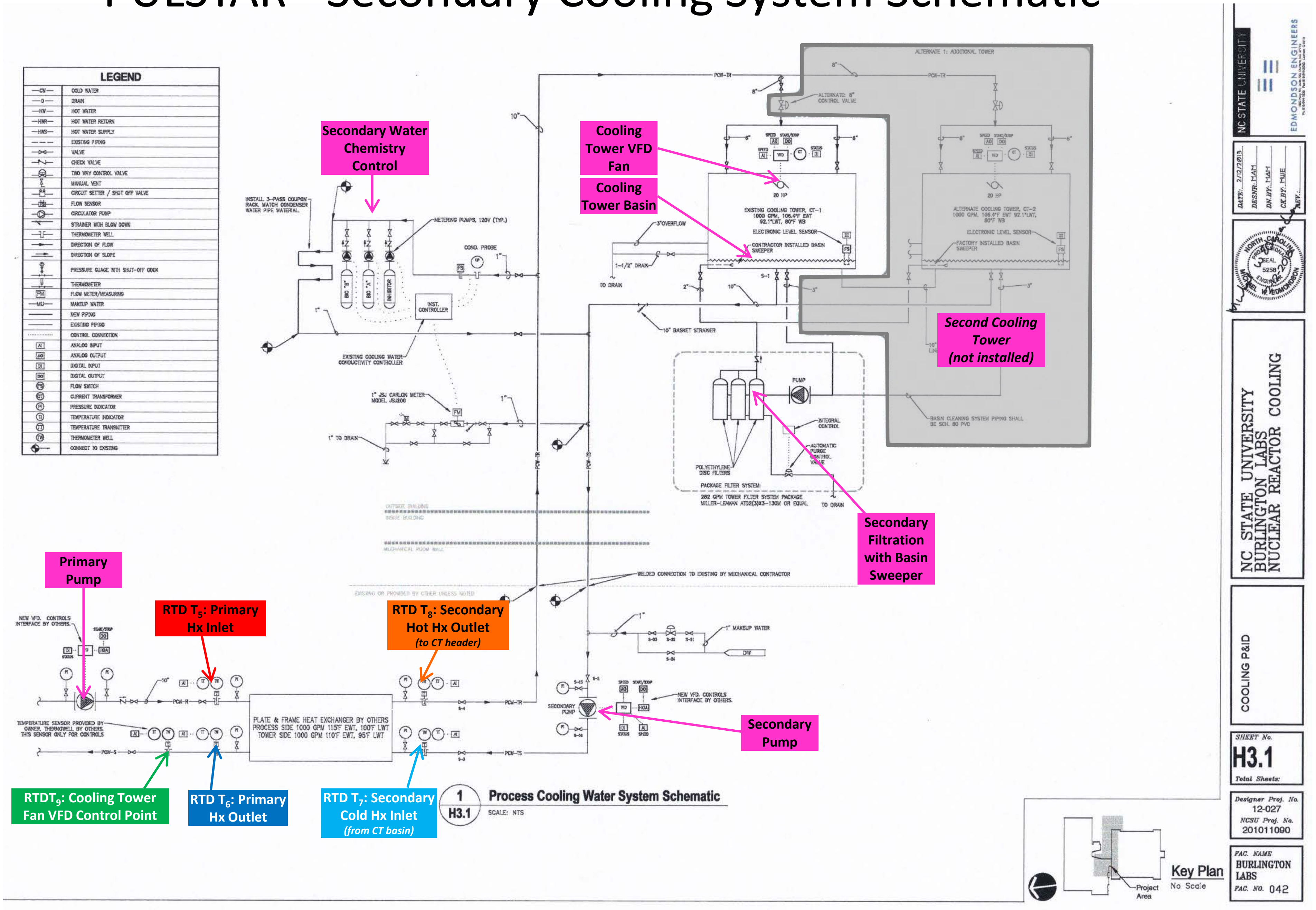
Heat Exchanger:

- 126 Plates – 0.5mm thick, 304 Stainless Steel
- 500 GPM Primary Flow Rate
- 700 GPM Secondary Flow Rate
- 2 MW Current Capacity; expandable.
- Design Pressure – 150 PSI
- Max Temp – 200°F
- 4.74 PSI drop at 1000 GPM

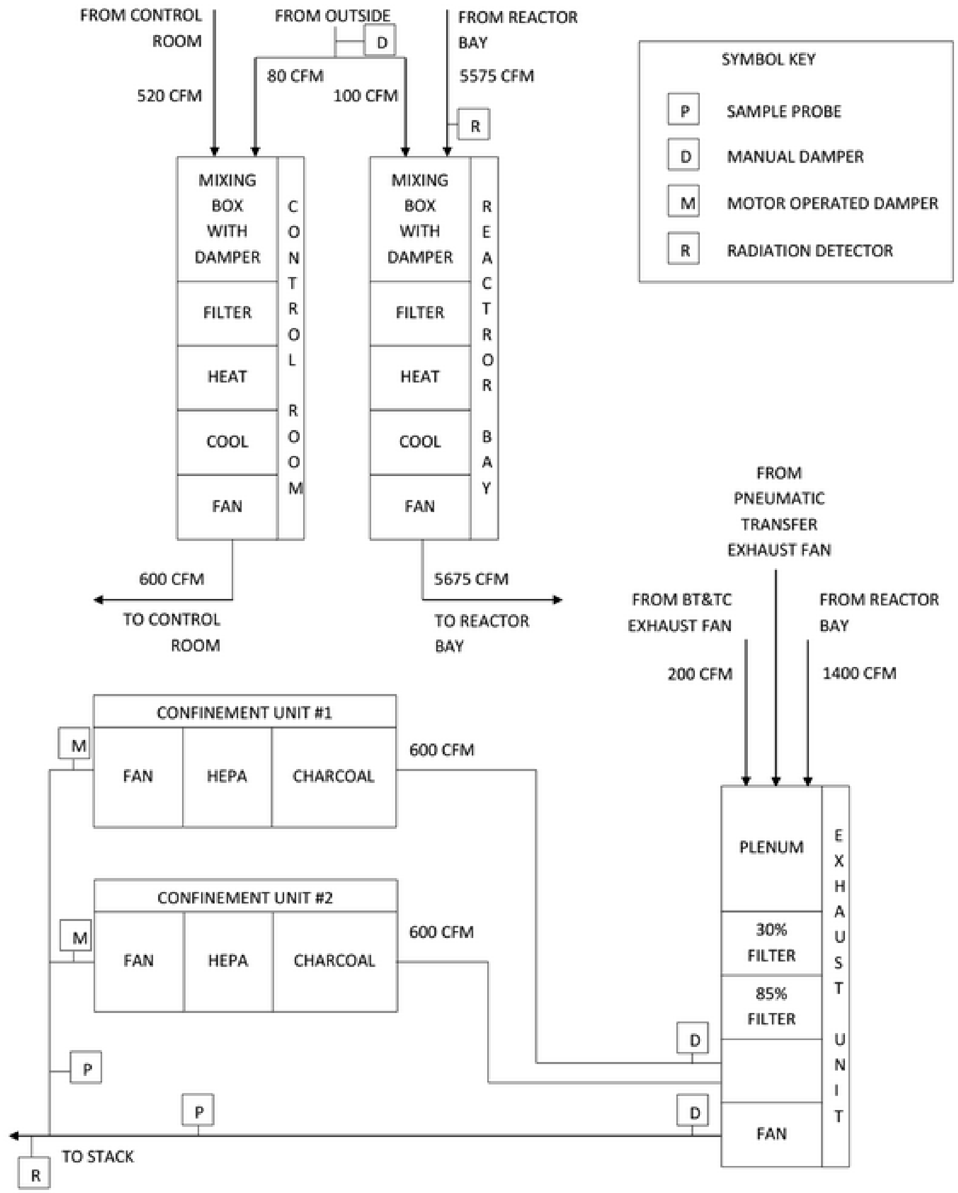
Nitrogen-16 Delay Tanks



PULSTAR - Secondary Cooling System Schematic



PULSTAR Confinement & HVAC Systems



PULSTAR Radiation Monitoring and EVAC System